

Problem 113. A positive integer is called *squarefree* if it has no square divisors larger than 1. For any positive integer n , define

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 p_2 \cdots p_k \text{ for some pairwise distinct primes } p_1, p_2, \dots, p_k, \\ 0 & \text{if } n \text{ is not squarefree.} \end{cases}$$

Let $x_1, x_2, \dots$ be a sequence of positive integers and let $y_n = \sum_{j|n} x_j$ for each positive integer n . Prove that

$$x_n = \sum_{j|n} \mu(j) y_{\frac{n}{j}}$$

for each positive integer n .

Problem 114. In a non-isosceles acute triangle ABC , D is the midpoint of BC . The points E and F lie on AC and AB , respectively, and the circumcircles of $\triangle CDE$ and $\triangle AEF$ intersect at P on AD . The interior angle bisector of $\angle FPE$ in $\triangle EFP$ intersects the line EF at Q .

Prove that the tangent line to the circumcircle of $\triangle AQP$ at A is perpendicular to BC .

Problem 115. Let $P(x)$ be a non-constant polynomial with real coefficients such that all of its roots are real numbers. Suppose that there exists a polynomial $Q(x)$ with real coefficients such that

$$(P(x))^2 = P(Q(x)) \quad \text{for all real numbers } x.$$

Prove that all the roots of $P(x)$ are equal.

Problem 116. For positive integer k , let

$$R_n = \begin{cases} \{-k, -(k-1), \dots, -1, 1, \dots, k-1, k\} & \text{for } n = 2k, \\ \{-k, -(k-1), \dots, -1, 0, 1, \dots, k-1, k\} & \text{for } n = 2k+1. \end{cases}$$

A device consists of several balls and red or white ropes connecting some pairs of balls. A *labelling* is a colouring of each ball by one of the elements of R_n . We say that a labelling is

- *good* if colours of any two connected balls are different;
- *sensitive* if the colours of any two balls connected by white rope are different and the sum of colours of any two balls connected by red rope is not equal to 0.

Let $n \geq 3$ be a fixed integer. Suppose that any device which has a good labelling by R_n has also a sensitive labelling by R_m . Find the smallest possible value of m as a function of n .